KoFLEURS-R: Improving Text and Audio Consistency in the Korean FLEURS-R Corpus

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***Abstract*—We present KoFLEURS-R, a refined version of the Korean portion of FLEURS-R that improves text-audio consistency at the recording level. FLEURS-R improves the acoustic quality of FLEURS recordings but retains the recording scripts, which do not always match what was actually spoken. In particular, parenthetical expressions may be spoken or omitted depending on the speaker, resulting in transcription mismatches. To address this issue, we propose a script-constrained refinement method that generates transcription candidates from the original text and selects the most appropriate candidate using acoustic evidence from two Whisper models, a Korean wav2vec 2.0 model, and a connectionist temporal classification (CTC) forced aligner. Evaluation on 189 human-verified recordings shows that KoFLEURS-R reduces the phonetic-key-normalized micro character error rate (CER) from 1.54% to 0.21%, an 86.4% relative reduction. The method outperforms both the removal of all parenthetical content and unconstrained automatic speech recognition (ASR), and correctly classifies 33 of 34 manually verified parenthetical occurrences (97.1%). The results suggest that script-constrained refinement can reduce transcript mismatches without replacing the script with error-prone ASR output. The refined metadata and the refinement pipeline will be released.**

***Index Terms*—Korean speech corpus, text-to-speech, text-audio consistency, FLEURS-R**

I. Introduction

Text-to-speech (TTS) models learn the correspondence between text and speech from paired data, so the text of each training pair must reflect what is actually spoken. Mismatched pairs teach the model incorrect correspondences and can degrade synthesis quality. To address this issue, large-scale speech corpora have employed forced alignment [1], automatic speech recognition (ASR) [2], [3], and connectionist temporal classification (CTC)-based alignment scores [4], [5] to identify and filter out mismatched text-audio pairs. Filtering, however, discards recordings that could be kept if the text were corrected instead.

FLEURS [6] is a multilingual read-speech corpus in which n-way parallel sentences from a machine translation benchmark are read aloud in 102 languages. FLEURS-R [7] restores the audio quality of FLEURS using Miipher [8], making the recordings suitable for speech generation tasks. However, FLEURS-R inherits the text of FLEURS unchanged; this text is the script given to the speakers rather than a transcription of what they said. Consequently, text-audio mismatches arise whenever speakers deviate from the scripts. Although FLEURS validated recordings against their corresponding sentences during data collection [6], and FLEURS-R used speech recognition to screen for restoration artifacts [7], neither revises the text to reflect the actual spoken content.

Such mismatches cannot be resolved by text-based rules alone. For example, parenthetical content in a recording script may be read aloud by one speaker but omitted by another. Retaining the content would introduce unspoken words into the text of the latter recording, whereas removing it would omit spoken words from the former. Even recordings that share a script may therefore require different text, and each recording must be examined on its own acoustic evidence.

In this paper, we present KoFLEURS-R, a refined version of the Korean portion of FLEURS-R in which the text of each recording is made consistent with its audio. Rather than replacing the original text with ASR output, our method selects from a closed set of candidates derived from the original sentence, so recognition errors cannot enter the text. Each parenthetical expression is treated as a slot that is either omitted or read aloud, and each number and acronym as a slot over its possible readings; the latter determines only the spoken form, and the written form is kept. For each recording, we choose slot options using phonetic edit distances to hypotheses from two Whisper models [9] and a Korean wav2vec 2.0 model [10], [11]. A CTC-based forced aligner [5], [12] supplies additional acoustic scores for ambiguous readings. We also provide a per-recording reliability score: the phonetic error rate of the refined text against the Whisper hypotheses, averaged over the two models, with verbalization differences excluded. The score measures agreement with the recognizers rather than a calibrated probability of transcription correctness.

The main contributions of this work are as follows:

- We characterize text-audio mismatches in FLEURS-R and show that recording scripts can differ from the spoken content.
- We propose a recording-level refinement method that selects from a closed set of script-derived candidates using acoustic evidence and provides a reliability score for each recording.
- We build KoFLEURS-R from the Korean portion of FLEURS-R, evaluate it against human-checked transcriptions, and will release the dataset and pipeline.

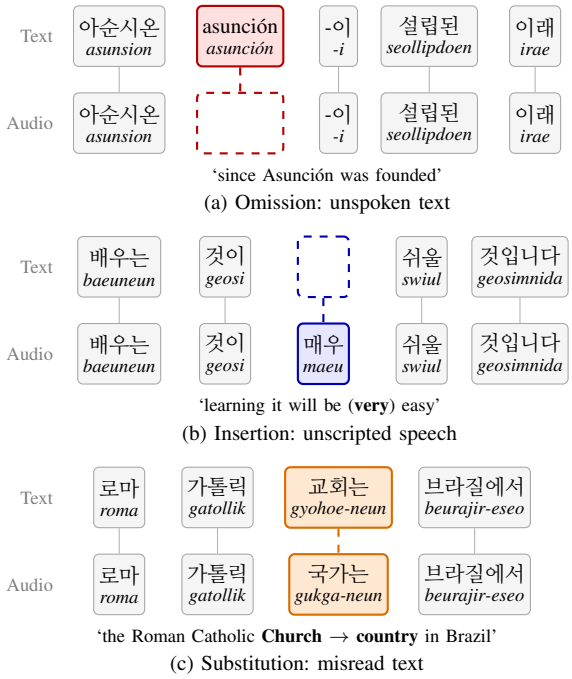


Fig. 1. Examples of omissions, insertions, and substitutions in Korean text-audio pairs, all confirmed by listening. Boxes show the script (top) and the spoken content (bottom); dashed links indicate mismatches. In (c), another speaker of the same script reads 교회는 as written.

II. Text-Audio Mismatches in FLEURS-R

A. Korean Subset of FLEURS-R

The Korean portion of FLEURS-R (`ko_kr`) contains 2,908 recordings of 1,742 sentences (2,302 train, 226 dev, and 380 test recordings; 10.0 h), each read by one to three speakers. Each recording is associated with two text fields: `raw_transcription`, the original translated sentence before normalization, and `transcription`, its normalized form, which serves as the reference transcript in the corpus [6]. Both fields are inherited unchanged from FLEURS and are shared by all recordings of the same sentence. We therefore use `transcription` as the text to be refined.

B. Types of Text-Audio Mismatches

Text-audio mismatches can be classified into three types (Fig. 1): *omission*, *insertion*, and *substitution*. Omission occurs when script content is not spoken; it frequently involves parenthetical content, which FLEURS normalization retains after removing the parentheses. Insertion occurs when speech contains content absent from the script, such as words added by the speaker. Substitution occurs when script content is spoken differently, typically as a result of a reading error. For instance, of the two speakers who read the script in Fig. 1(c), one reads 교회는 (*gyohoe-neun*, 'church') as written, whereas the other says 국가는 (*gukga-neun*, 'country'). Differences in the verbalization of numbers, acronyms, and foreign words are not considered mismatches, as these elements can have multiple valid readings [13], [14]. We further group these types into two categories according to whether they can be anticipated from the script: the omission of parenthetical content, which can be located in the script, is a *predictable* mismatch, whereas other omissions, insertions, and substitutions are *unpredictable* mismatches.

C. Recording-Dependent Parenthetical Content

The presence of a parenthetical expression in a script does not determine whether a speaker pronounced it. For example, a Latin-script gloss may be omitted, while a Hangul explanation may be spoken as part of the sentence; neither behavior is universal. In our corpus analysis (Section IV-A), 40 parenthetical expressions received different decisions across recordings of the same source sentence, which shows that the decision must be made per recording.

III. Proposed Method

As discussed in Section II, omissions of parenthetical content are predictable, whereas other omissions, insertions, and substitutions are unpredictable. Our method therefore treats the two categories separately: it corrects predictable mismatches through recording-level candidate selection (Section III-A) and quantifies the remaining unpredictable mismatches with a reliability score (Section III-B).

A. Recording-Level Candidate Selection

We generate candidate texts from `raw_transcription`, in which the parentheses are still present. Each parenthetical expression defines a slot with two basic options, omitted or read, and a read expression may have several alternative readings. For example, "크레바스(crevasse)" may be read as "크레바스" (*keurebaseu*), with the gloss omitted, or as "크레바스 크레바스" (*keurebaseu keurebaseu*)[1], with the gloss read in Hangul. Similarly, we define slots for numbers and acronyms, which may have multiple readings; for instance, "3" may be read as "삼" (*sam*) or "셋" (*set*). A candidate is formed by selecting one option for each slot. We then convert each candidate into a spoken form, spelling out numbers and acronyms in Hangul, for comparison with the speech.

To identify the candidate that best matches each recording, we use recognition hypotheses from two Whisper models [9] and a Korean wav2vec 2.0 model [10], [11]. Parenthetical decisions use only the Whisper hypotheses, with ties resolved in favor of omission; the wav2vec 2.0 model supplies additional evidence for number and acronym readings. Before comparing candidates and hypotheses, we map both to a coarse phonetic key ϕ that represents Hangul and Latin letters with a common symbol set. Following the minimum Bayes-risk principle [15], [16], option o for slot i is scored by its average edit distance to the available hypotheses:

$$J_i(o) = \frac{1}{|\mathcal{E}_i|} \sum_{m \in \mathcal{E}_i} d\big(\phi(c_{i \leftarrow o}), \phi(h_m)\big), \tag{1}$$

where $c_{i \leftarrow o}$ denotes the candidate with option o assigned to slot i, h_m is the hypothesis of model m, $\mathcal{E}_i$ is the set of models

[1]Italics in parentheses give the Revised Romanization of Korean.

whose hypotheses are used for slot i, and d is the Levenshtein distance [17] over the utterance. We update each slot with its lowest-scoring option in turn; this coordinate-wise search is not guaranteed to find the globally optimal combination. For ambiguous acronym readings, we additionally use the CTC likelihood from a forced aligner [5], [12].

Once the slot options have been selected, we add two text fields to each recording, keeping the original fields unchanged. The `refined_text` field is derived from `transcription` by removing the parenthetical content selected as omitted, while retaining the original written forms of numbers and acronyms. The `spoken_text` field contains the spoken form of the selected candidate and can be used as text input for speech synthesis. In addition, we report a signed margin for each parenthetical expression, derived from the recognition hypotheses and the CTC likelihood, that indicates the relative support for reading versus omission.

B. Reliability Score

Although candidate selection addresses predictable mismatches, unpredictable mismatches cannot be resolved from the script alone. Following the common practice of assessing transcript quality with ASR error rates [2], [18], we quantify the remaining disagreement between the refined text and the recognition hypotheses using a reliability score:

$$R = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \frac{e_m}{N}, \tag{2}$$

where $\mathcal{M}$ is the set of Whisper models, N is the number of phonetic symbols in the refined text after numbers and acronyms are converted to their default readings, and e_m is the number of edit operations in the Levenshtein alignment [17] between the refined text and hypothesis h_m. To avoid penalizing differences in verbalization, variations involving numbers, acronyms, and Latin-script words are excluded unless the corresponding span is judged as not spoken, which is determined by whether the proportion of deleted symbols exceeds a threshold τ.[2] The resulting score is stored in the `reliability_score` field; lower scores indicate greater agreement with the recognition hypotheses, not necessarily greater accuracy.

Since R depends on recognition hypotheses, we also provide a complementary metric derived directly from the speech: a CTC posterior-based acoustic support score inspired by goodness-of-pronunciation (GOP) measures [19]. For each aligned frame, it subtracts the maximum token log posterior from the log posterior of the aligned character. The difference is close to zero when the aligned character is the most probable token and becomes negative when another token has a higher posterior probability. The word-level average summarizes the agreement between the alignment and the CTC predictions; it is not a calibrated phone-level GOP score. The `min_gop` field stores the minimum of these averages across words.

[2]In this work, τ is set to 0.5; it can be adjusted to control how strictly a span is judged as not spoken.

TABLE I
CONSTRUCTION STATISTICS OF KOFLEURS-R BY SPLIT. PARENTHETICAL COUNTS ARE AUTOMATIC DECISIONS; RECORDINGS WITH CHANGED TEXT INCLUDE QUOTATION-MARK REPAIRS.

	Train	Dev	Test	All
Recordings	2,302	226	380	2,908
Hours	7.9	0.8	1.3	10.0
Parenthetical expressions	291	31	71	393
Judged omitted	183	22	57	262
Judged spoken	108	9	14	131
Recordings with omission	147	18	46	211
Recordings with changed text	289	26	59	374
Quotation-mark repairs	150	8	19	177

IV. EXPERIMENTS

A. Construction of KoFLEURS-R

We applied the proposed refinement pipeline to all 2,908 recordings of the Korean FLEURS-R corpus, including the train, development, and test splits. Recognition hypotheses from Whisper large-v3 and large-v3-turbo were obtained with beam search (beam size 5, temperature 0); the wav2vec 2.0 model and the forced aligner are public checkpoints.[3] The pipeline selected a script-derived candidate independently for each recording, retaining parenthetical content when it was judged to have been spoken and removing it otherwise. We also repaired malformed quotation marks in 177 recordings without changing their spoken content.

Table I summarizes the resulting corpus. Of the 393 occurrences of parenthetical expressions, 262 were judged omitted and 131 spoken. In total, 211 recordings contain at least one omitted expression, and 374 have revised text, including the quotation-mark repairs. Among the parenthetical expressions occurring in multiple recordings, 40 received different spoken/omitted decisions across recordings of the same source sentence. These decisions are automatic, not ground truth; their accuracy is examined against human judgments in Section IV-D.

B. Evaluation Protocol

To assess text-audio consistency, one author manually reviewed 190 of the 380 test recordings by listening to the audio and correcting the provided transcription where necessary. These recordings form two of four sessions into which the test split was randomly divided, stratified by the presence of parentheses, digits, and Latin letters. One recording marked as uncertain was excluded, leaving 189 recordings of 165 distinct source sentences. The human references were produced by editing the original scripts rather than by blind transcription; because the scripts were visible during annotation, the evaluation may favor script-preserving approaches (see Section V).

We compared four text sources: the original FLEURS-R transcription, a rule-based baseline that removes every parenthetical expression (Remove-all), the unconstrained output of

[3]`kresnik/wav2vec2-large-xlsr-korean` and `MahmoudAshraf/mms-300m-1130-forced-aligner` on Hugging Face.

TABLE II
PHONETIC-KEY-NORMALIZED CER (%) OF THE FOUR TEXT SOURCES AGAINST HUMAN REFERENCES ON 189 MANUALLY CHECKED TEST RECORDINGS.

Text source	Macro CER ↓	Micro CER ↓
Original FLEURS-R	2.17	1.54
Remove-all	0.96	0.87
Whisper large-v3	1.02	0.96
KoFLEURS-R (ours)	**0.24**	**0.21**

TABLE III
PARENTHETICAL-CONTENT DECISION PERFORMANCE ON 34 HUMAN-VERIFIED OCCURRENCES (26 OMITTED, EIGHT SPOKEN). OMISSION IS THE POSITIVE CLASS.

Method	Accuracy (%) ↑	F1 (%) ↑
Remove-all	76.5	86.7
Type-majority	61.8	71.1
KoFLEURS-R (ours)	**97.1**	**98.1**

Whisper large-v3, and the proposed KoFLEURS-R transcription. All sources were scored against the same human references after the phonetic-key mapping ϕ of Section III, which removes spacing and punctuation differences and normalizes alternative written realizations for phonetic comparison. We report micro CER, the total number of edit errors divided by the total reference length, and macro CER, the mean of the recording-level error rates. Both are edit rates over the study-specific phonetic key rather than strict orthographic CER or phoneme error rate.

C. Transcript Refinement Accuracy

Table II reports the accuracy of the four text sources. KoFLEURS-R achieved a micro CER of 0.21% and a macro CER of 0.24%, compared with 1.54% and 2.17%, respectively, for the original FLEURS-R transcription, an 86.4% relative reduction in micro CER. The method also outperformed Remove-all (0.87% micro CER) and Whisper large-v3 (0.96% micro CER). At the recording level, refinement improved the CER of 22 recordings relative to the original transcription, worsened one, and left 166 unchanged.

The two baselines test two different explanations. Remove-all tests whether the improvement can be obtained by simply deleting parenthetical text, whereas Whisper large-v3 tests whether replacing the script with an unrestricted ASR hypothesis is sufficient. KoFLEURS-R had lower CER than either baseline, which supports making recording-dependent decisions while retaining the script content outside the editable slots.

To assess the uncertainty of these paired comparisons, we performed 10,000 cluster-bootstrap resamples [20], [21] at the source-sentence level, accounting for multiple recordings of the same sentence. The reductions in micro CER relative to the original transcription, Remove-all, and Whisper were 1.33, 0.66, and 0.75 percentage points, with 95% confidence intervals of [0.65, 2.23], [0.13, 1.36], and [0.52, 1.02], respectively; none included zero. Sentence-level sign-flip randomization tests [22] yielded $p < 0.001$ against the original transcription and Whisper, and $p = 0.014$ against Remove-all.

D. Recording-Level Parenthetical Decisions

The proposed method is designed to distinguish parenthetical content that was actually spoken from content omitted by the speaker. We therefore evaluated its decisions on the 34 parenthetical occurrences in the human-reviewed recordings, of which 26 were omitted and eight were spoken. As baselines, we use Remove-all, which corresponds to always predicting omission, and Type-majority, which applies to each kind of parenthetical content (e.g., Latin-script gloss or Hangul explanation) the decision made most often for that kind in the automatic decisions on the whole corpus.

As shown in Table III, KoFLEURS-R correctly classified 33 of the 34 occurrences, identifying all 26 omissions while incorrectly omitting one of the eight spoken expressions; its precision and recall for the omitted class were 96.3% and 100%, respectively. Remove-all correctly classified only 26 occurrences (76.5% accuracy), and the paired difference between the two was significant under the exact McNemar test [23] ($p = 0.016$). These results support making the decision per recording: parenthetical text cannot be handled reliably by unconditional deletion or by a rule that depends only on its text type.

E. Comparison with Unconstrained ASR

To examine the benefit of constraining corrections to the source script, we compared recording-level exact matches to the human references for Whisper large-v3 and KoFLEURS-R under the same phonetic-key normalization. Whisper was incorrect while KoFLEURS-R was correct for 55 of the 189 recordings; the reverse occurred in only two. Both were correct on 119 recordings and both incorrect on 13. This asymmetry is consistent with the goal of retaining the reliable words of the original script instead of propagating recognition errors, such as misrecognized proper names, into the refined text; Whisper is also known to hallucinate content [24]. It does not imply that all 55 cases were actively corrected by our pipeline; some were already correct in the original script and were simply preserved.

V. DISCUSSION

On the manually checked Korean FLEURS-R subset, script-constrained refinement produced fewer text-audio mismatches than either Remove-all or direct ASR transcription. Its main advantage is conservative editing: the system changes only the parts of the script for which it has candidate alternatives and preserves the rest. This also limits its scope: insertions, substitutions, and unexpected omissions cannot be corrected when the relevant candidate is absent.

The 189 human references were prepared by a single annotator with the original scripts visible. This may favor methods that retain script text, and no inter-annotator agreement was measured. In 17 recordings, both Whisper models agreed with

each other but not with the reference; re-listening to one of them revealed a missed insertion, which we corrected. These 17 recordings are among the 55 in Section IV-E, so this bias makes the comparison with the original text conservative but may overstate the advantage over Whisper. The reliability score only weakly identified the 15 recordings that still differ from the reference after refinement (ROC-AUC 0.64, 95% CI 0.49–0.80), largely because eight of them differ only in written form, which the score does not penalize; computed on the original transcription, however, it identified the 27 mismatched recordings with an ROC-AUC of 0.99. The decisions for the train and development splits have not been verified by listening. Moreover, our normalized CER does not capture all orthographic differences, and we have not evaluated downstream TTS performance. Future work will ablate the individual components, include independent blind transcriptions, and measure the effect of the corrected metadata on speech synthesis. Extending the method to other FLEURS-R languages will require language-specific candidate generation and phonetic normalization; multilingual CTC aligners such as MMS [5] provide a starting point, but verbalization rules for numbers and acronyms differ across languages [13].

VI. Conclusion

We introduced KoFLEURS-R, a recording-level transcript refinement of the Korean FLEURS-R corpus. Using script-derived candidates and acoustic evidence, our method reduced the phonetic-key-normalized micro CER from 1.54% to 0.21% on 189 manually checked recordings. It correctly classified 33 of 34 parenthetical occurrences and outperformed both Remove-all and unconstrained ASR. These findings support correcting predictable transcript mismatches while retaining the remaining script content. The effect on downstream TTS quality remains to be tested. The refined metadata and the pipeline will be released.

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